

SHARORAH, Saudi Arabia

WMO: 411360

Lat: 17.467N

Lon: 47.121E

Elev: 2363

StdP: 13.48

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.8	51.2	13.8	12.1	70.2	17.7	14.6	73.5	21.1	67.9	18.9	67.1	2.9	80	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	109.6	70.5	108.9	70.4	107.6	70.2	81.2	103.2	77.5	100.3	74.9	100.1	8.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	144.0	98.1	70.1	121.1	95.5	66.4	106.3	92.2	47.3	102.2	43.1	100.4	40.2	100.0	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.5	15.5	DB	41.2	112.0	4.1	1.3	38.2	113.0	35.9	113.7	33.6	114.5	30.6	115.4	(4)
(5)			WB	32.7	78.4	4.1	4.5	29.8	81.6	27.4	84.2	25.1	86.8	22.1	90.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	66.6	72.4	79.2	85.9	91.2	93.8	95.6	95.1	90.8	82.0	74.3	67.7	(6)
(7)		DBStd	11.19	6.18	6.33	5.31	4.21	2.82	2.65	2.22	2.60	3.34	3.92	4.38	5.85	(7)
(8)		HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	114	58	14	0	0	0	0	0	0	0	0	2	40	(9)
(10)		CDD50	12024	514	627	905	1077	1278	1314	1415	1399	1223	993	730	549	(10)
(11)		CDD65	6662	106	221	441	627	813	864	950	934	773	528	281	124	(11)
(12)		CDH74	106620	1284	2707	5745	9425	13712	15309	17013	16476	12952	7464	3185	1348	(12)
(13)	CDH80	70581	433	1265	3133	5880	9422	11067	12573	12034	8861	4222	1282	409	(13)	
(14)	Wind	WSAvg	7.2	7.2	7.8	8.4	8.4	7.6	6.4	6.9	7.6	6.6	6.4	6.8	6.7	(14)
(15)	Precipitation	PrecAvg	2.5	0.0	0.1	0.6	0.6	0.2	0.2	0.2	0.1	0.1	0.0	0.0	(15)	
(16)		PrecMax	10.5	0.4	1.3	4.1	4.4	1.0	3.5	2.0	2.6	3.2	1.1	0.4	0.0	(16)
(17)		PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.3	0.1	0.3	1.1	1.1	0.3	0.7	0.4	0.5	0.6	0.2	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.4	97.1	101.8	106.2	108.0	111.0	111.4	110.4	108.0	100.7	93.4	91.4	(19)
(20)			MCWB	62.5	63.4	65.2	70.0	67.8	70.6	71.0	71.4	70.0	67.3	66.5	64.5	(20)
(21)		2%	DB	88.0	93.8	98.7	102.6	106.5	109.3	109.7	109.3	106.2	98.7	91.2	87.8	(21)
(22)			MCWB	62.4	63.4	65.5	68.1	68.5	69.6	70.6	71.6	69.7	66.3	64.4	63.4	(22)
(23)		5%	DB	84.6	91.3	96.6	100.7	105.5	107.8	109.0	107.8	105.1	96.9	89.4	84.5	(23)
(24)			MCWB	60.7	63.7	65.0	67.1	68.8	68.8	70.5	71.6	69.1	65.3	63.7	61.1	(24)
(25)		10%	DB	81.8	88.0	93.5	98.9	103.8	106.2	107.6	107.0	103.6	94.9	87.5	81.4	(25)
(26)			MCWB	59.5	62.0	64.6	66.4	68.7	68.2	70.4	71.4	68.5	64.9	62.7	59.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	70.2	72.3	74.7	78.5	80.3	85.8	85.1	80.5	74.1	70.9	68.3	(27)
(28)			MCDWB	87.0	90.3	91.7	98.0	100.9	104.2	106.1	104.1	102.3	91.5	88.4	85.9	(28)
(29)		2%	WB	64.7	67.1	69.8	72.5	75.7	75.6	80.9	81.4	74.1	70.8	67.4	65.2	(29)
(30)			MCDWB	84.0	88.7	90.3	96.4	100.4	98.8	103.4	103.2	100.0	90.2	85.5	84.4	(30)
(31)		5%	WB	62.4	65.0	68.1	70.9	73.6	73.1	76.2	77.9	71.5	68.1	65.6	62.9	(31)
(32)			MCDWB	81.6	87.3	88.8	95.8	99.3	101.8	101.0	101.3	101.1	93.0	85.4	81.6	(32)
(33)		10%	WB	60.4	62.8	66.3	69.2	71.4	71.0	73.1	74.6	69.7	66.3	63.8	60.9	(33)
(34)			MCDWB	78.8	85.1	88.8	94.9	98.7	103.4	101.7	100.3	100.7	92.3	84.2	79.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	25.6	26.6	26.3	25.3	25.2	26.4	25.3	24.2	25.5	25.3	24.7	24.8	(35)
(36)			MCDBR	29.8	31.0	29.6	27.5	27.3	27.5	26.2	25.1	26.2	26.2	25.8	27.1	(36)
(37)			MCWBR	15.0	14.6	13.2	12.3	12.4	12.0	11.4	11.0	11.6	12.1	12.8	13.0	(37)
(38)		5% WB	MCDWB	27.5	28.1	25.8	24.4	24.2	25.4	24.0	23.5	25.7	24.7	24.4	25.4	(38)
(39)			MCWBR	15.1	14.9	13.5	12.5	13.8	13.9	14.1	13.6	13.4	12.8	13.1	13.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.429	0.465	0.523	0.527	0.583	0.672	0.715	0.677	0.561	0.438	0.410	0.405	(40)
(41)		taud		1.941	1.839	1.720	1.746	1.615	1.444	1.428	1.494	1.662	1.925	2.009	2.005	(41)
(42)		Ebn at Noon		268	265	254	253	236	214	205	215	240	269	272	271	(42)
(43)		Edh at Noon		56	66	76	75	84	99	101	95	80	59	52	51	(43)
(44)	All-Sky Solar Radiation	RadAvg		1716	1953	2174	2315	2397	2359	2249	2226	2218	2102	1823	1670	(44)
(45)		RadStd		71	63	74	82	62	75	75	58	26	47	40	82	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	+144
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page